Unit 3: SOLVED PROBLEMS

1.Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

2.Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

3.Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

4.Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

5.Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

6.A uniformly distributed variable ranges from 10 to 50. Find P(20<X<35).

7.A normal distribution has mean 100 and standard deviation 15. Find the Z-score for X=130.

8.For the same normal distribution, find P(X<130).

9.Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

10.Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

1. Find Mean, Median, Mode and Range

Given Data

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange in ascending order

12, 15, 15, 15, 18, 20, 22, 25

Number of observations

Step 2: Mean

Formula

Calculate the sum

Therefore,

Mean = 17.75

Step 3: Median

Since there are 8 observations (even),

Middle positions are

and

4th value = 15

5th value = 18

Median

Median = 16.5

Step 4: Mode

Value	Frequency
12	1
15	3
18	1
20	1
22	1
25	1

The highest frequency is 3.

Therefore,

Mode = 15

Step 5: Range

Formula

Maximum = 25

Minimum = 12

Range = 13

Final Answer

- **Mean = 17.75**
- **Median = 16.5**
- **Mode = 15**
- **Range = 13**

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2. Construct a Frequency Distribution

Given Data

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Count Frequencies

2 appears

2,2,2

Frequency = 3

3 appears

3,3,3

Frequency = 3

4 appears

4,4,4

Frequency = 3

5 appears

5,5

Frequency = 2

6 appears

6

Frequency = 1

Verification

which matches the total observations.

Value	Frequency
2	3
3	3
4	3
5	2
6	1

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3. Find Q1, Q2, Q3 and IQR

Given Data

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Already arranged.

Number of observations

Step 1: Median (Q2)

Since n is even,

Middle observations

5th = 15

6th = 18

Step 2: Lower Half

5, 8, 10, 12, 15

Middle value = 10

Therefore,

Step 3: Upper Half

18,20,22,25,30

Middle value =22

Therefore,

Step 4: IQR

Formula – Q3-Q1

Final Answer

Q1 = 10

Q2 =16.5

Q3 =22

IQR =12

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4. Detect Outliers Using IQR Method

Given Data

10,12,13,15,16,17,18,20,45

n=9

Median =16

Lower Half

10,12,13,15

Since there are four values,

Upper Half

17,18,20,45

IQR

Lower Fence

First calculate

Then

Upper Fence

Compare Values

Observation	Within Range?
10	Yes
12	Yes
13	Yes
15	Yes
16	Yes
17	Yes
18	Yes
20	Yes
45	No

Therefore,

45 is an outlier.

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5. Determine Skewness

Given Data

5,6,7,7,8,8,9,20

Mean

Sum

Mean

Median

Middle observations

4th =7

5th =8

Since

The distribution is positively (right) skewed.

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6. Uniform Distribution

Given

Minimum =10

Maximum =50

Required interval

20 to35

Formula

Length of interval

Total interval

Probability

Answer = 0.375 or 37.5%

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7. Z-score

Given

Mean =100

Standard deviation =15

X =130

Formula

Substitute values

Answer: Z = 2

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8. Find P(X<130)

From Question 7,

Using the standard normal table,

Therefore,

or

97.72%

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9. Compare Mean and Median Before and After Outlier Treatment

Original Data

10,12,13,15,16,17,18,20,45

Before Removing Outlier

Sum

Mean

Median =16

Remove 45

New data

10,12,13,15,16,17,18,20

Sum

Mean

Median

Comparison

Measure	Before	After
Mean	18.44	15.125
Median	16	15.5

Conclusion: The mean decreases considerably after removing the outlier, whereas the median changes only slightly, demonstrating that the median is less affected by extreme values.

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10. Complete EDA

Dataset

45,50,55,60,65,70,72,75,80,95

Number of observations

Mean

Sum

Mean

Median

Middle values

65 and70

Mode

Every value appears once.

No mode

Range

Quartiles

Lower half

45,50,55,60,65

Q1 =55

Upper half

70,72,75,80,95

Q3 =75

IQR

Outlier Limits

Lower

Upper

Since every value lies between 25 and 105, there are no outliers.

Summary

Measure	Value
Mean	66.7
Median	67.5
Mode	No Mode
Minimum	45
Maximum	95
Range	50
Q1	55
Q2	67.5
Q3	75
IQR	20
Lower Limit	25
Upper Limit	105
Outliers	None
Distribution	Approximately symmetric
Skewness	Slightly negative (Mean < Median)